Experiment : 5

Date:

CLASS DIAGRAM AND TIME SEQUENCE DIAGRAM – ATM SYSTEM

Aim:

To design and analyze the UML Class Diagram and Time Sequence Diagram for an ATM Banking System in order to understand its structural components and interaction flow during transaction processing.

Introduction:

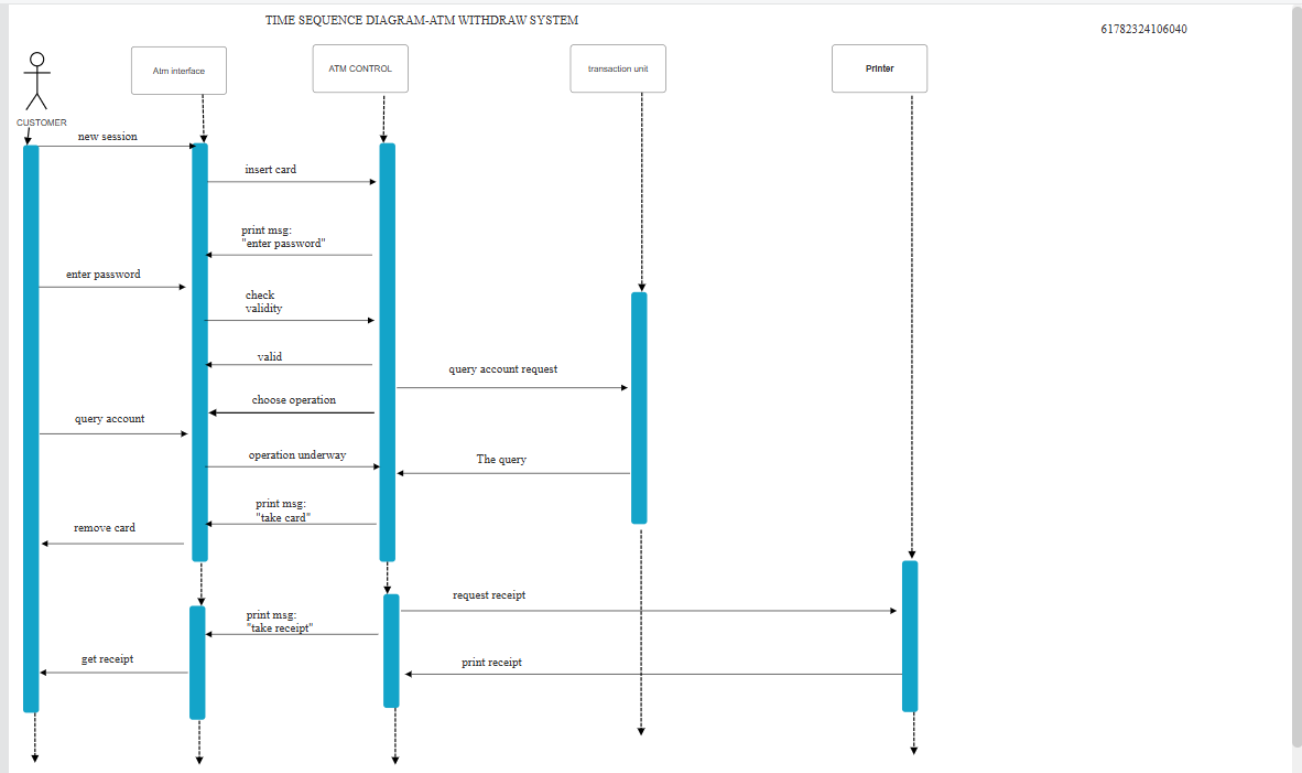
Unified Modeling Language (UML) is a standardized modeling framework used to design and represent software systems visually. It enables developers to understand both structural and behavior aspects of applications.

In this experiment, UML diagrams are designed for an **ATM Banking System**. The Class Diagram represents the static architecture of the ATM system including classes such as Bank, Customer, Account, ATM, and Transaction. The Time Sequence Diagram represents the dynamic flow of events occurring during an ATM withdrawal process.

These diagrams collectively provide a complete understanding of how the ATM system is structured and how it operates during runtime.

Class Diagram : ATM Banking System

The Class Diagram represents the static structure of the ATM system. It shows various classes, their attributes, operations, and relationships such as association and inheritance.



Classes Identified in the ATM System:

1. **Bank**
   - Attributes: code, address
   - Method: getAccount()
2. **Customer**
   - Attributes: name, address, dob, cardNumber, pin
   - Method: verifyPassword()
3. **Account**
   - Attributes: accountNumber, balance
   - Methods: deposit(), withdraw()
4. **Checking Account** (inherits from Account)
   - Contains specific features unique to checking accounts
5. **Saving Account** (inherits from Account)
   - Contains specific features unique to savings accounts
6. **ATM**
   - Attributes: location, managed_by
   - Methods: withdraw(), deposit(), checkBalance()

7. **ATM Transaction**
    - Attributes: transactionId, date, type, amount, postBalance
    - Method: modify()

Relationships in the Class Diagram:

- A **Bank** manages multiple **ATMs**.
- A **Customer** owns an **Account**.
- **Account** is the parent class of **Checking Account** and **Saving Account** (Inheritance).
- The **ATM** interacts with **Account** to perform transactions.
- Each transaction is recorded in **ATM Transaction**.

Importance of the ATM Class Diagram

• Provides clear structural representation of the ATM system
• Identifies relationships between banking entities
• Supports object-oriented design principles
• Improves system maintainability
• Helps in planning real-time banking applications

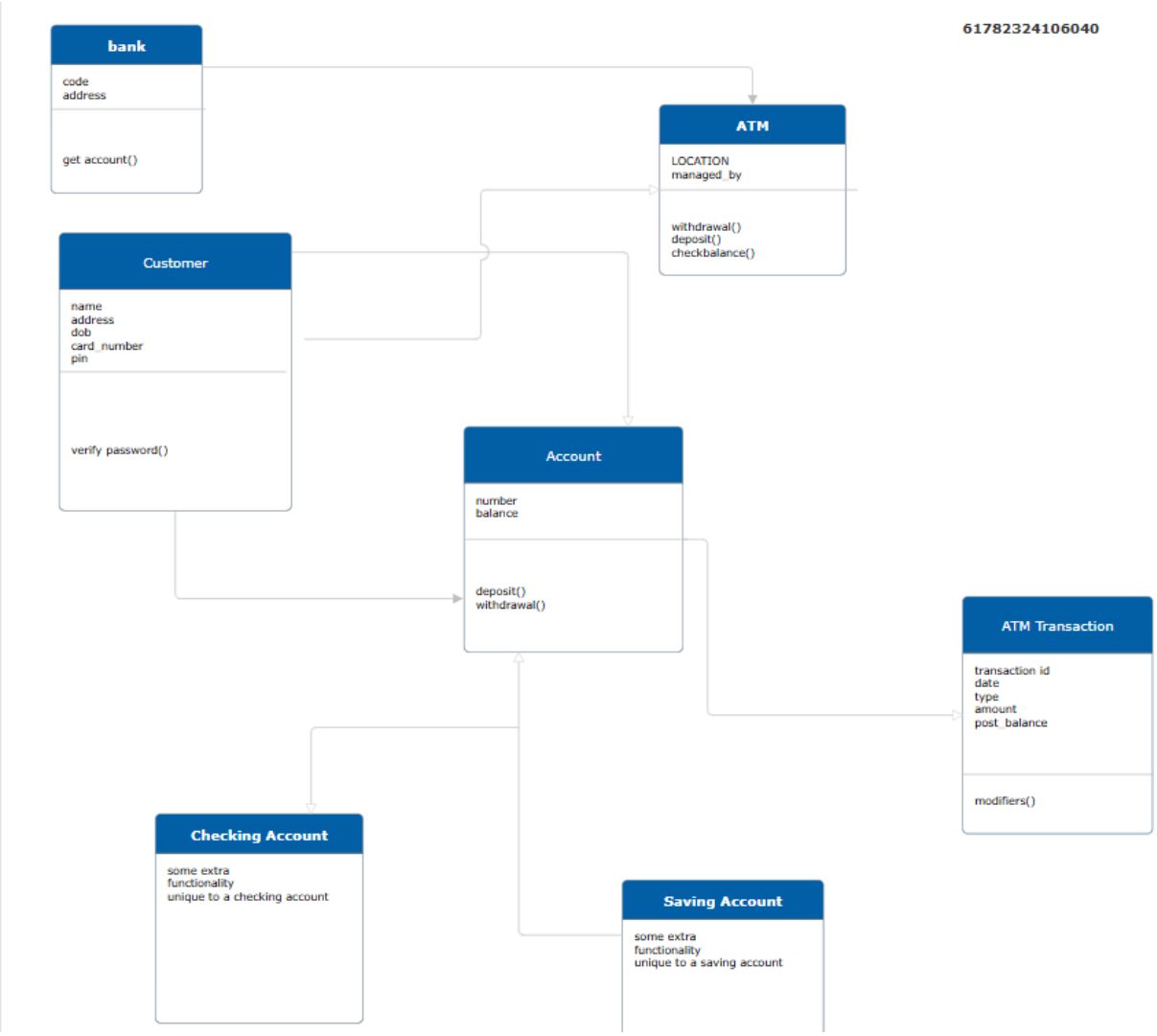
Applications of the Class Diagram

1. ATM Software Development
2. Banking System Architecture Design
3. Financial Transaction Modelling
4. Object-Oriented Implementation
5. System Documentation

The ATM Class Diagram acts as a blueprint that defines how banking components are organized and interconnected before actual implementation.

Time Sequence Diagram : ATM Withdrawal System

The Time Sequence Diagram represents the dynamic behavior of the ATM during a withdrawal transaction. It shows how messages are exchanged between system components over time.



Objects/Lifelines Identified:

- Customer
- ATM Interface
- ATM Control
- Transaction Unit
- Printer

Sequence of Interaction (ATM Withdrawal Process):

1. Customer starts a new session.
2. Customer inserts ATM card.
3. System prompts to enter password.
4. Customer enters password.

5. ATM Control verifies the password.
6. If valid, user selects operation (withdraw).
7. ATM sends account query to Transaction Unit.
8. Transaction Unit processes the request.
9. Operation is completed.
10. ATM displays message to remove card.
11. Customer removes card.
12. Receipt request is sent to Printer.
13. Printer prints receipt.
14. Customer collects receipt.

Purpose of ATM Time Sequence Diagram

1. Shows step-by-step transaction processing
2. Illustrates interaction between ATM components
3. Helps analyze authentication and validation flow
4. Ensures correct execution sequence
5. Supports real-time transaction modelling

Key Elements Used

• Lifelines – Represent system components
• Messages – Indicate communication between components
• Activation Bars – Show processing duration
• Time Flow – Top to bottom execution

Functional Importance

• Explains ATM workflow clearly
• Identifies validation and transaction stages
• Helps detect logical sequencing errors
• Supports secure banking operations
• Useful in financial system design

Advantages of UML Modelling in ATM System

• Simplifies complex banking processes
• Enhances system clarity
• Improves documentation quality
• Enables early error detection
• Reduces development risks

Conclusion:

Thus, the UML Class Diagram and Time Sequence Diagram for the ATM Banking System were successfully designed and analyzed. The Class Diagram clearly represented the structural components such as Bank, Customer, Account, ATM, and Transaction, while the Time Sequence Diagram illustrated the interaction flow during a withdrawal transaction.

This experiment strengthened understanding of object-oriented system design and interaction modelling. It also emphasized the importance of UML diagrams in developing secure, scalable, and well-structured banking software systems.